

Chlorophyll Extraction

Overview

Extract chlorophyll of each set of Arabidopsis groups after plant experimental trials for readout data and analysis.

A. Materials

- a. Agar plates with plants
- b. Tweezer
- c. Tissue paper
- d. Weighing plate
- e. 15ml centrifuge tube
- f. Marker
- g. Stainless steel beads
- h. Acetone
- i. Vortex machine
- j. OD machine
- k. Plate with wells to load samples

B. Procedure

1. Using a tweezer, lift and pick up all arabidopsis plants simultaneously from the agar plate in a single motion
 - a. Careful not to hurt the plants by squeezing the tweezer
 - b. Lift from below the leaf area
1. Put the plants on a tissue paper, lightly press to remove the agar sticking to the plants
 - a. Careful not to press the leaf areas too hard
2. Place all plants in a weighing plate, weigh the fresh weight (fw) of arabidopsis (a plate at a time)
3. Pick up the plants and put them in a 15ml centrifuge tube, take care that the entire plant and the roots need to go in too
4. Label the (1)group concentration and (2)respective fresh weight (fw) on the microtube
5. Repeat these steps for as many times as the sets of arabidopsis you are testing

6. Place 10 stainless steel beads in each of the centrifuge tubes, place in -80 degrees C for 10 minutes
7. Dilute acetone to 80%, add 4 mL to each tube
8. Wait for the chlorophyll to extract, and can use a vortex or shake vigorously to crush the plants to speed up the process (done when plants are no longer green.)
9. Label what samples are in each well before loading ***very important
10. Add the liquid to the circular 96 well, 200 μ L in each well, 3 wells for each sample.
 - a. Blank OD for 200 μ L 80% acetone
11. ELISA reader, insert the 96 well, run the results through the equation below.
 - a. Make sure to select the correct wavelength
 - i. A663
 - ii. A645

*Total chlorophyll content equation(mg/g):

$$(20.2 * OD645 + 8.02 * OD663) V / (1000 * fw)$$

V=2ml

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